

Kinetic studies to investigate lipoprotein metabolism

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Abstract. To develop novel strategies for the prevention and treatment of dyslipidaemia, it is essential to understand the pathophysiology of dyslipoproteinaemia in humans. Lipoprotein metabolism is a complex system in which abnormal concentrations of various lipoprotein particles can result from altera-

tions in their rates of production, conversion and/or catabolism. Traditional methods that measure plasma lipoprotein concentrations only provide static estimates of lipoprotein metabolism and hence limited mechanistic information. By contrast, the use of tracers labelled with stable isotopes and mathematical modelling provides a powerful tool for probing lipid and lipoprotein kinetics *in vivo* and furthering understanding of the pathogenesis of dyslipoproteinaemia.

Keywords: kinetics, lipoproteins, metabolism, stable isotopes, very low-density lipoproteins.

Introduction

The function of lipoprotein particles is to transport hydrophobic lipid molecules, which are not water soluble, in the blood. The general structure of a lipoprotein includes a core consisting of triglycerides and/or cholesteryl esters surrounded by a surface monolayer of phospholipids, unesterified cholesterol and specific proteins [1]. The protein components of the lipoprotein are known as apolipoproteins [1]. The different apolipoproteins serve as cofactors for enzymes and ligands for receptors. The handling of lipoproteins in the body is referred to as lipoprotein metabolism.

Disorders of lipoprotein metabolism are often linked to cardiovascular disease (CVD) [2]. In order to prevent and treat CVD, it is necessary to characterize these metabolic disorders to fully understand their cause. The most common way to do this is by measuring plasma lipid or apolipoprotein concentrations. However, abnormal concentrations of lipids and apolipoproteins can result from changes in the production, conversion or catabolism of lipoprotein particles. Therefore, although static measurements and functional assays are important techniques, it is necessary to study the true unit of function (the integrated metabolic pathway) to understand the complexity of lipoprotein metabolism [3–5].

In contrast to static measurements, kinetic experiments are of great importance in lipid research because they further the understanding of lipid metabolism *in vivo* and help to explain the pathophysiology of lipid disorders in humans [3–5]. At present, because of species specificity, no valid animal model can efficiently replace human studies to explore lipid metabolism, and the use of radioactive tracers in such studies is restricted. Therefore, stable-isotope tracer kinetic studies have become an important tool of translational research to achieve a quantitative understanding of the dynamics of metabolic processes *in vivo*. The aim of this review is to describe this methodology and illustrate how the approach has furthered our understanding of the pathogenesis of disorders of human lipoprotein metabolism.

Lipoprotein metabolism

Fats absorbed in the intestine are packaged into large, triglyceride-rich particles known as chylomicrons (Fig. 1). These lipoproteins undergo lipolysis (removal of triglycerides) to form chylomicron remnants, which are taken up by the liver. The liver can also secrete triglyceride-rich lipoproteins known as very low-density lipoproteins (VLDLs). Lipoprotein kinetic studies have shown that VLDLs are metabolically heterogeneous, with accumulating evidence demonstrating that both the production and

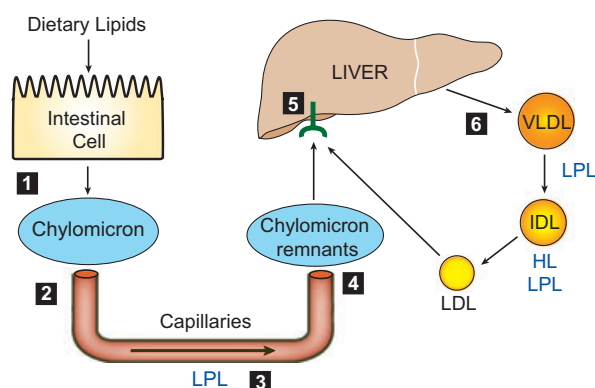


Fig. 1 Metabolism of apolipoprotein B (apoB)-containing lipoproteins. Dietary lipids are packed into chylomicrons (1). Once in the plasma (2), most triglyceride molecules in chylomicrons are hydrolysed by the action of lipoprotein lipase (LPL) (3) resulting in smaller chylomicron remnants (4). The remnants are cleared from the circulation by the liver (5). LPL and hepatic lipase (HL) convert liver-derived very low-density lipoproteins to smaller intermediate-density lipoprotein (IDL) and LDL (6), which are removed from the circulation by the liver.

catabolism of large triglyceride-rich VLDL₁ and smaller cholesterol-rich VLDL₂ are regulated independently of each other. Variations in plasma triglyceride concentrations are mainly accounted for by differences in VLDL₁. Following lipolysis, these particles can be converted to low-density lipoprotein (LDL) or taken up by the liver. The LDL formed is catabolized mainly by the liver or other tissues via the LDL receptor. Increased plasma concentration of LDL is a major risk factor for CVD. High-density lipoprotein (HDL) is synthesized by both the liver and the intestine. HDL picks up lipid and protein constituents from chylomicrons and VLDL as these particles undergo lipolysis. The plasma concentration of HDL is inversely associated with the development of CVD.

Principles of tracer methodology

To study the behaviour of an endogenous molecule, the *tracee*, the same molecule, but labelled (the *tracer*), is introduced into the system (usually via the bloodstream) [4, 6]. This process is termed *exogenous labelling*. *Endogenous labelling* occurs when a labelled precursor of the molecule of interest is used to label this molecule (e.g., infusion of a labelled amino acid to label a protein). Ideally, the tracer can easily be detected and quantified, has the same kinetic behaviour as the tracee, and does not perturb the system. Usually, kinetic studies are performed in a steady state, where the rates of input and output for a

given unlabelled tracee substance are equal and time invariant. Thus, the information provided by the tracer reflects the behaviour of the tracee [7]. At various times, the amount of tracer is quantified to provide a kinetic curve. Then a mathematical model is constructed to extract all the information contained in the kinetic curve. By fitting a model to the data, it is possible to calculate the parameters of the model that characterize the flux of molecules between kinetically homogeneous pools. For example, it is possible to calculate the production, conversion or catabolism of lipoprotein particles, information that cannot be obtained by static measurements alone. Lipoprotein kinetic studies can also be modelled under non-steady-state conditions [8], but this requires certain complex assumptions to be made about production and/or catabolic rates.

The term stable isotope refers to a non-radioactive isotope of a given atom that is less abundant in a molecule within a biological system than the lightest naturally occurring isotope. Stable isotopes commonly used as metabolic tracers include [²H₃]-leucine for apolipoprotein turn-over; [²H₅]-glycerol and [¹³C]-palmitate for VLDL-triglyceride metabolism; [1,2-¹³C]-acetate and [¹³C]-palmitate for fatty acid metabolism and hepatic *de novo* lipogenesis; ²H₂O for *de novo* lipogenesis in the liver; [¹³C₅]-cholesterol and [²H₆]-cholesterol for intestinal cholesterol absorption; and [¹³C]-acetate for assessing HDL-mediated reverse cholesterol transport. Stable isotope tracers are much safer than radioactive tracers for both the study subject and the investigator. Recent advances in mass-spectrometry technology permit accurate measurement of stable isotopes in smaller samples and at lower concentrations [7, 9].

Tracer administration

A tracer can be administered intravenously as either a bolus injection or a primed constant infusion (i.e., a constant infusion given immediately after a priming dose). The bolus administration of tracer displays superior dynamics compared with the primed constant infusion, because the enrichment curves (the tracer/tracee ratios) after a bolus injection correspond to the impulse response of the system. It is therefore suitable to study components of lipoprotein metabolism with a slow rate of turnover. Another advantage of bolus administration is that it facilitates the determination of newly synthesized particles, as the intracellular precursor enrichment is greater at the start of the study. Practically, the bolus infusion

is also most convenient for both subjects and investigators.

An advantage of using the primed constant infusion approach, on the other hand, is that the analysis of tracer data is relatively simple. However, it requires a longer time to achieve a plateau of isotopic enrichment in lipoproteins with a low rate of turnover. This must be considered in clinical studies because assumptions about the plateau of isotopic enrichment have to be made, for practical reasons, in studies lasting less than approximately 12 h. The primed constant infusion is therefore suitable for lipoproteins with rapid turnover, such as VLDL. Long infusion protocols also have the disadvantage of tracer recycling, although this can, to some extent, be accounted for in modelling the tracer data.

Data analysis: multicompartment models

When a stable isotope tracer is injected, it is incorporated into the molecules under investigation. The fraction of labelled molecules can be estimated by measuring the enrichment of the stable isotope (tracer/tracee ratio). The tracer enables the dynamic fate of the latter to be traced through its metabolic pathways. The movement of material in the system can be described by a set of differential equations, thereby allowing quantitative information on the system to be estimated.

The tracer data generated can be analysed using several different methodologies. One approach to analyse VLDL kinetics is to inject a bolus of tracer and determine the subsequent monoexponential slope of the decline in plasma VLDL concentration. A disadvantage of this approach is that it can underestimate the true VLDL turnover rate because it does not account for recycling of the injected bolus of tracer [10]. Multicompartment modelling is a superior method to dissect the complexities of lipoprotein metabolism, and has been widely applied to systems in which material is transferred over time between compartments connected in a specific structure to permit the movement of material amongst the compartments.

Each compartment is assumed to be a homogenous entity within which the entities being modelled are equivalent. For instance, the compartments may represent different types of lipoprotein particles that are kinetically homogeneous and distinct from other material in the system. Very often, the data can be described by more than one model. To ensure that the best model is selected, it is necessary to carefully

examine the fitting of the kinetic curve, to determine the precision of the parameter estimates, and to perform statistical tests to compare results obtained with different models. However, the complexity of a multicompartment model is usually a compromise for what is practically possible. A very simple model may not adequately describe the kinetic heterogeneity present within the system. A model that is too complex, on the other hand, will not be supported by experimental data and, hence, will have little predictive value. Furthermore, even if the development of models is based on experimental data, several assumptions are required in order to derive the model that is to be used. Thus, mathematical models do not determine the kinetics of lipids directly; rather, they derive an indirect approximation.

The software SAAM (Epsilon Group, VA) has become the cornerstone for modelling lipoprotein kinetics studies. The SAAM II program was developed recently, and is frequently used to aid the design and analysis of lipoprotein tracer data using compartmental models [11, 12].

A conceptually simpler approach than multicompartmental modelling is administration of a 'true' VLDL-triglyceride tracer (i.e., whole VLDL particles containing a lipid tracer such as triglycerides containing a stable or unstable isotope), with subsequent calculation of VLDL-triglyceride kinetics [13]. In theory, as precursor kinetics are not involved, study protocols of shorter duration are sufficient to determine kinetic parameters of interest, and tracer recycling problems are circumvented. Such true tracers may be produced *in vivo* by administration of a precursor tracer followed by isolation and reinfusion [13]. The method is intuitively very appealing and elegant, but tracer production involving plasmapheresis (i.e., the removal, treatment and return of circulating blood plasma) is laborious and therefore not practical in all laboratories. Sorensen *et al.* [14] have developed an alternative method to produce a true VLDL-triglyceride tracer based on *ex vivo* labelling with subsequent reinfusion. In addition to calculation of VLDL-triglyceride kinetic parameters of interest, the method allows the study of the metabolic fate of VLDL-triglycerides, by tracing the associated fatty acids.

Historically, the use of mathematical modelling to describe the metabolic pathways of lipid and lipoprotein metabolism was pioneered by Drs. Berman and Zech in the 1970s [15, 16]. Their work was highly influenced by scientists working in the 1950s and 1960s, who developed approaches to

study complex molecular systems by quantitative modelling. Today, this approach has become widespread and is known as systems biology.

Lipid transport modelling

Several studies have been conducted to analyse VLDL–triglyceride turnover kinetics using stable isotopically labelled glycerol or palmitate tracers and mathematical modelling (Fig. 2); however, VLDL subclasses were not analysed and VLDL–apolipoprotein B (apoB) was not included in the models [10, 17, 18]. To enhance our understanding of the pathways leading to VLDL₁ and VLDL₂ and of the metabolic fate of these particles, we developed a multicompartmental mathematical model that allows the kinetics of triglycerides and apoB100 in VLDL₁ and VLDL₂ to be simultaneously assessed after a bolus injection of glycerol and leucine [19]. Tracer/tracee curves of leucine and glycerol in VLDL₁ and VLDL₂ were analysed to determine kinetic parameters using mathematical modelling.

Analysis of the enrichment curves shows that enrichment rapidly increases, which indicates that influx is much greater than efflux (Fig. 3). At the peak, either the influx starts to decrease, or the efflux balances the influx. During the period of rapid decay, efflux predominates. After the peak of VLDL₁, labelled material enters VLDL₂ both directly from the liver and from VLDL₁. By simultaneously modelling apoB and triglyceride kinetics, it was possible to determine the triglyceride/apoB ratio of newly produced VLDL₁ and VLDL₂ particles and follow in detail the transfer and removal of lipids [19].

The data were analysed with two linear compartmental models. The model can be envisioned as a two-layer model, connected at certain points, and the model can be envisioned as a two-layer model, consisting of two linear compartmental models

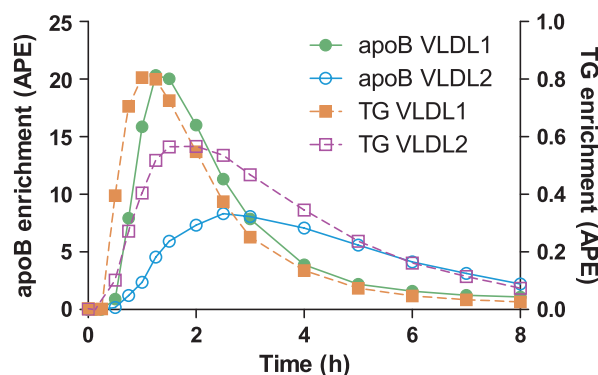


Fig. 3 Typical atom per excess (APE) enrichment curves for triglyceride and apoB. Triglyceride (TG)–very low-density lipoproteins (VLDL₁) (■, dashed/dotted line) and TG–VLDL₂ (□, dotted line) are plotted on the right axis and the apoB–VLDL₁ (●, solid line) and apoB–VLDL₂ (○, dashed line) on the left axis. The two VLDL₁ curves are similar in shape to each other, but the two VLDL₂ curves show differences with regard to the initial slope and the rate of clearance. The difference in delay times between TG and apoB is clearly shown.

connected at certain points and is based on the apoB model originally described by Packard *et al.* [20] which has been used in several studies [20–26]. Basically, the model consists of four components: plasma leucine, plasma glycerol, the assembly of lipoprotein and lipoprotein plasma kinetics (Fig. 4). Plasma kinetics are modelled by a four-compartment hydrolysis chain, where the apoB and triglyceride kinetics are coupled at the transfer between compartments. Removal from triglyceride compartments consists of removal of both whole particles and triglycerides.

Disturbed lipid metabolism in subjects with type 2 diabetes and the metabolic syndrome

A hallmark of insulin resistance is a dyslipidaemia characterized by increased plasma triglycerides and

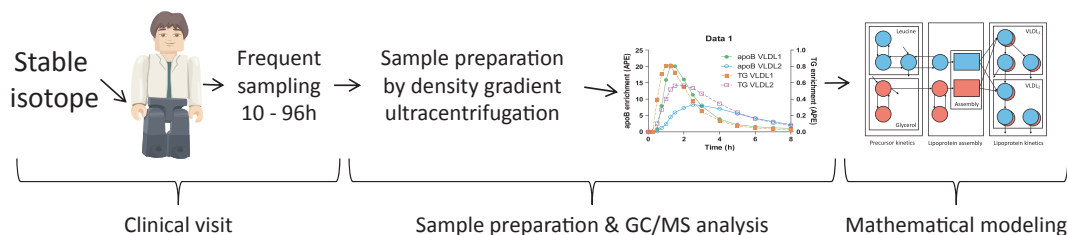


Fig. 2 Outline of kinetic tracer studies. A kinetic study includes three major steps: (i) clinical visit protocol (tracer infusion protocol); (ii) isolation of lipoproteins using density gradient ultracentrifugation and tracer/tracee enrichment analysis by gas chromatography mass-spectrometry (GC/MS); and (iii) mathematical modelling.

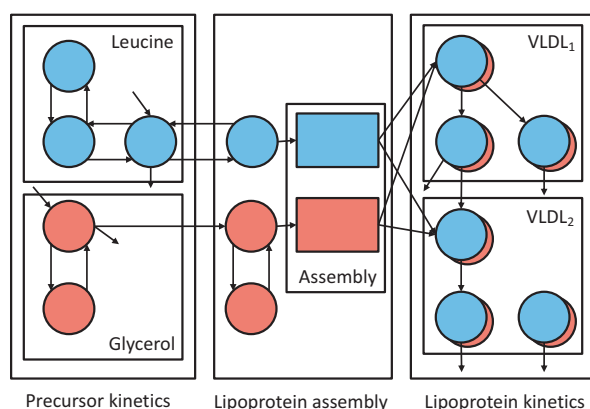


Fig. 4 Compartmental model of very low-density lipoproteins (VLDL) apoB and triglyceride kinetics. The model includes separate modules for leucine and glycerol. Plasma leucine kinetics are modelled using a four-compartment system that drives the synthesis and secretion of apoB into VLDL₁ and VLDL₂. Plasma glycerol kinetics are modelled using a two-compartment system connected to fast and slow pathways for triglyceride (TG) synthesis. Plasma apoB and triglyceride kinetics are modelled using a four-compartment hydrolysis chain, in which the kinetics of apoB and triglycerides are coupled. For each apoB compartment, there is an equivalent compartment for triglycerides. Triglycerides hydrolysed from VLDL particles are represented by the dashed arrows, and particles lost from the plasma space are represented by the solid arrows. See Adiels *et al.* [19] for additional model details.

low concentrations of HDL cholesterol [27]. More recently, recognized features are small dense LDL (sdLDL) and excessive postprandial lipaemia (defined as a rise in triglyceride-rich lipoproteins after eating) [28]. Today, it is established that the different components of this diabetic dyslipidaemia (sometimes also referred to as atherogenic dyslipidaemia or the lipid triad) are not isolated abnormalities but are metabolically closely linked to each other [29].

Patients with diabetic dyslipidaemia have a significant increased risk of developing CVD. Furthermore, the prognosis for CVD is much worse in the presence of type 2 diabetes. However why the diabetic dyslipidaemia is atherogenic remains unclear. Many studies have shown that sdLDL is a strong independent predictor of CVD [30]. Normally, liver cells secrete small VLDL particles (VLDL₂). Intravascular lipolysis of these lipoproteins gives rise to normal-size LDL that remains in the circulation for ~2 days. However, when triglyceride levels are high, VLDL₁ particles are initially converted to large LDL particles. These large LDL particles are converted to sdLDL

with a retention time of ~5 days by cholesterol ester transport protein and hepatic lipase, which are both increased in type 2 diabetes [28]. In addition to this, increased levels of VLDL₁ alter the composition of HDL, ultimately leading to an increased catabolism of these particles [31]. Thus, elevation of VLDL₁ in diabetic dyslipidaemia leads to the generation of sdLDL and increased risk of CVD. Of importance, these atherogenic lipid abnormalities appear years before type 2 diabetes is diagnosed and may explain the 'ticking clock' hypothesis of CVD [28]. Kinetic studies have been vital in understanding the complexity of the disturbed lipoprotein metabolism in diabetic dyslipidaemia (Fig. 5).

Knowledge of the pathophysiology of dyslipidaemia from kinetic studies

To further our understanding of diabetic dyslipidaemia, we analysed which features of type 2 diabetes

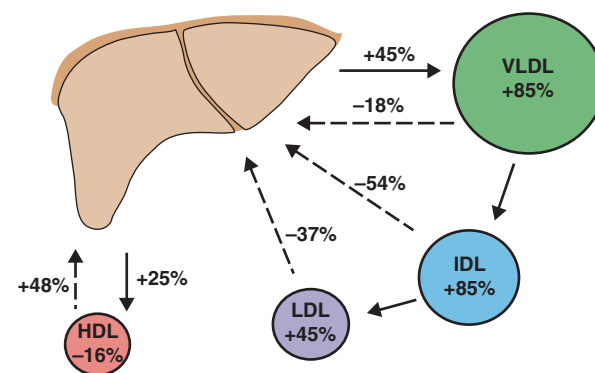


Fig. 5 The dysregulation of apoB and apoAI metabolism in subjects with the metabolic syndrome. ApoB-100 is the major structural protein of very low-density lipoproteins (VLDL), IDL and LDL. It is a large protein that is required for the assembly and secretion of apoB-containing lipoproteins. It is also a ligand for LDL receptor-mediated endocytosis of LDL and, possibly, VLDL and IDL. Subjects with the metabolic syndrome have significantly increased production (+45% for VLDL, solid line) and reduced catabolism of apoB100 (-18%, -54% and -37% for VLDL, IDL and LDL respectively, dashed lines). This results in an increase in the concentration of VLDL, IDL and LDL particles. ApoAI is the major structural protein associated with high-density lipoprotein (HDL) particles. It is important in the activation of enzymes involved in the maturation of HDL particles. In addition, a low HDL cholesterol concentration is observed in these subjects. This is primarily because of hypercatabolism of its major structural protein apoAI (+48%, dashed line), which is somewhat compensated by increased production (+25%, solid line). The figure is based on studies by Watts *et al.* [50–52].

and insulin resistance correlate with VLDL₁ production, and revealed strong correlations with plasma glucose that are not apparent in the normal range of glucose concentration [32]. By extending our study to monitor liver, intra-abdominal and subcutaneous fat as well as adiponectin, we showed that fasting insulin, plasma glucose, intra-abdominal and liver fat and the homeostasis model of assessing insulin resistance (HOMA-IR) are predictors of VLDL₁-apoB and VLDL₁-triglyceride production [33]. However, in a multiple regression analysis, only liver fat and plasma glucose remained significant [33]. Moreover, the key predictors of liver fat are intra-abdominal fat, adiponectin and plasma glucose [33]. Further studies have demonstrated that liver fat content is closely related to triglyceride secretion in different settings and populations [34, 35] and is a better predictor of triglyceride secretion than intra-abdominal fat [36, 37].

Non-alcoholic fatty liver disease (NAFLD) is defined as fat accumulation in the liver that exceeds 5–10% of liver weight in individuals who do not consume significant amounts of alcohol [38]. Recent data show that NAFLD is strongly associated with type 2 diabetes, obesity and hyperlipidaemia [39–46]. We tested the relationship between liver fat and VLDL₁ suppression amongst subjects with a broad range of liver fat content [47]. From this study, we confirmed that liver fat predicts baseline VLDL₁ production [33], and also that liver fat is associated with a lack of VLDL₁ suppression in response to insulin. Indeed, insulin downregulates VLDL₁ secretion in subjects with low liver fat levels, but fails to suppress VLDL₁ secretion in subjects with high liver fat levels, resulting in overproduction of VLDL₁ [47]. The reason for this lack of VLDL₁ suppression is not known. Insulin suppresses the non-esterified fatty acid (NEFA) pool to a similar extent regardless of liver fat level [47], and thus, the high VLDL₁ production in individuals with high levels of liver fat must be facilitated either by using a greater portion of systemic NEFA or recruiting other sources of triglycerides, such as from hepatic stores. The data suggest that liver fat content in obesity may be a better marker of metabolic derangement and CVD risk than visceral obesity *per se*.

Recently, we performed kinetic studies with lipoproteins to elucidate why some, but not all, obese subjects develop dyslipidaemia [34]. Specifically, we tested whether hypertriglyceridaemia in obese men with a similar body mass index and waist circumference is caused solely by increased hepatic secretion of VLDL induced by increased liver fat. Unexpectedly, our results showed that the hypertriglyceridaemia

can be explained by the combination of increased secretion and severely impaired clearance of triglyceride-rich VLDL₁ particles. The increased liver and subcutaneous abdominal fat are linked to increased secretion of VLDL₁ particles, whereas increased plasma levels of apoC-III are associated with impaired clearance in obese hypertriglyceridaemic subjects. These results provide new insights into the pathophysiology of dyslipidaemia in obesity.

There is much evidence that metabolic diseases primarily contribute to hypertriglyceridaemia by excessive secretion of triglycerides from the liver. It has been suggested that chronic elevated insulin levels, present in these states in parallel with insulin resistance, may contribute to the phenotype by increasing triglyceride biosynthesis. To test this hypothesis, a novel study was performed by Duvillard *et al.* [48] in patients with insulinoma. These patients have high insulin levels because of pancreatic overproduction but do not develop insulin resistance and glucose levels are below normal. The authors showed normal production of VLDL particles in patients with hyperinsulinaemia but without insulin resistance, despite an increase in serum insulin level that was similar in these subjects compared with those with insulin resistance. Thus, the findings support the notion that chronic endogenous hyperinsulinaemia alone is not responsible for the increased production of VLDL particles in patients with insulin resistance [48].

Physiological relevance of the integration of models

The interface between biological systems and nutritional factors represents the next level of complexity in metabolic research and requires the combination of multiple experimental and theoretical approaches. For example, it has long been known that the potential sources of fats contributing to fatty liver include peripheral fats stored in adipose tissue that flow to the liver by way of the plasma NEFA pool (pathway 1); fatty acids newly made within the liver through *de novo* lipogenesis (pathway 2); and dietary fatty acids, which can enter the liver through spillover into the plasma NEFA pool [13] (pathway 3) or through the uptake of intestinally derived chylomicron remnants (pathway 4). However, to clarify the relative contribution of the four sources, Donnelly *et al.* developed an elegant protocol using several different stable isotopes and liver biopsies to simultaneously measure and model all four pathways of fatty acid delivery to the liver in patients with suspected NAFLD. These quantitative metabolic data demonstrated for the

first time that both elevated peripheral fatty acid flux and *de novo* lipogenesis contribute to hepatic and lipoprotein fat in NAFLD [49].

The next challenge will be to develop bioanalytical and dynamic modelling tools for studies of systemic lipid metabolism in the context of nutritional interventions. New mathematical models must therefore be developed that include not only hepatic, but also intestinal lipid metabolism. Therefore, chylomicrons need to be included in the models. This is very complicated, explaining why quantitative nutritional studies require a suite of computational and analytical tools.

Conclusion

Lipid homeostasis is essential for human health and many if not most of today's major healthcare issues are related in some way to dysregulation of lipid metabolism. The use of stable isotopes and mass spectrometry in combination with the development of mathematical models, using a systems biology approach, has played a fundamental role in advancing our knowledge of metabolic pathways of lipid and lipoprotein metabolism *in vivo* by identifying covariates that are associated with normal and dysregulated lipid metabolism. Thus, the use of kinetic studies plays a central role in modern research, and through the combined use of mathematical models and experimental data, it is possible to obtain predictive models for simulations, such as for identification of drug targets or metabolic engineering strategies. However, kinetic studies are time consuming and expensive and require a high level of expertise. The success of kinetic studies relies on a multi-disciplinary team involving biochemists, physicians and biostatisticians, and therefore, these studies are best carried out in specialized research centres.

Conflict of interests

No conflict of interest was declared.

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